

РБНФ №1 (опис синтаксису всіма допустимими засобами РБНФ)	РБНФ №2 (опис формальної граматики засобами РБНФ)	Формальна граматика	Формальна граматика з специфікацією lookahead у правилах для LL(2)-аналізатора	Перевірка РБНФ №1 за допомогою коду <i>(помістити у файл "EBNF_N1.h")</i> */	Перевірка РБНФ №2 за допомогою коду <i>(помістити у файл "EBNF_N2.h")</i> */	Перевірки прототипу LL(2)-синтаксичного аналізатора (спеціальна структура) та прототипу лексичного аналізатора (регулярні вирази) за допомогою коду. Лексеми для синтаксичного аналізатора обробляються лексичним аналізатором, тому синтаксичний аналізатор не аналізує їх повторно (як показано в РБНФ). <i>(помістити у файл "LexicaByRegExAndSyntaxByLL2prototype.h")</i> УВАГА: при копіюванні зажайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
		G = {N, T, P, S}	G = {N, T, P, S}			
		S → program_rule	S → program_rule			
		N = { program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident_iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action_iteration, expression_or_cond_block_with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else_iteration, body_for_false_optional, cycle_begin_expression, cycle_end_expression, cycle_counter, cycle_counter_lr_init, cycle_counter_init, cycle_counter_last_value, cycle_body, fordownto_cycle, statement, statement_or_block_statements, block_statements, input_rule, argument_for_input, output_rule, statement_iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit_iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	N = { program_name, value_type, array_specify, declaration_element, array_specify_optional, other_declaration_ident, declaration, other_declaration_ident_iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action_optional, expression, binary_action_iteration, expression_or_cond_block_with_optional_assign, assign_to_right, assign_to_right_optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else_iteration, body_for_false_optional, cycle_begin_expression, cycle_end_expression, cycle_counter, cycle_counter_lr_init, cycle_counter_init, cycle_counter_last_value, cycle_body, fordownto_cycle, statement, statement_or_block_statements, block_statements, input_rule, argument_for_input, output_rule, statement_iteration, expression_optional, program_rule, declaration_optional, non_zero_digit, digit_iteration, digit, unsigned_value, value, sign_optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	#define NONTERMINALS program_name, \ value_type, \ array_specify, \ declaration_element, \ \ other_declaration_ident, \ declaration, \ \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ \ expression, \ \ expression_or_cond_block_with_optional_assign, \ assign_to_right, \ \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ \ cycle_begin_expression, \ cycle_end_expression, \ cycle_counter, \ cycle_counter_lr_init, \ cycle_counter_init, \ cycle_counter_last_value, \ cycle_body, \ fordownto_cycle, \ statement, \ statement_or_block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ \ program_rule, \ \ non_zero_digit, \ digit_iteration, \ digit, \ unsigned_value, \ value, \ \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	#define NONTERMINALS program_name, \ value_type, \ array_specify, \ declaration_element, \ array_specify_optional, \ other_declaration_ident, \ declaration, \ other_declaration_ident_iteration, \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ index_action_optional, \ expression, \ binary_action_iteration, \ expression_or_cond_block_with_optional_assign, \ assign_to_right, \ assign_to_right_optional, \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ false_cond_block_without_else_iteration, \ body_for_false_optional, \ cycle_begin_expression, \ cycle_end_expression, \ cycle_counter, \ cycle_counter_lr_init, \ cycle_counter_init, \ cycle_counter_last_value, \ cycle_body, \ fordownto_cycle, \ statement, \ statement_or_block_statements, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ statement_iteration, \ expression_optional, \ program_rule, \ declaration_optional, \ non_zero_digit, \ digit_iteration, \ digit, \ unsigned_value, \ value, \ sign_optional, \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	
		T = { "INTEGER16", " ", "NOT", "AND", "OR", "==" , "!=", "<=", ">=", "+", "-", "*", "DIV", "MOD",	T = { "INTEGER16", " ", "NOT", "AND", "OR", "==" , "!=", "<=", ">=", "+", "-", "*", "DIV", "MOD",	#define TOKENS \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ 	#define TOKENS \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ 	

				tokenBEGINBLOCK = "{" >> BOUNDARIES;	tokenBEGINBLOCK = "{" >> BOUNDARIES;	#define T_BEGIN_BLOCK_0 "{" #define T_BEGIN_BLOCK_1 "" #define T_BEGIN_BLOCK_2 "" #define T_BEGIN_BLOCK_3 ""
				tokenENDBLOCK = "}" >> BOUNDARIES;	tokenENDBLOCK = "}" >> BOUNDARIES;	#define T_END_BLOCK_0 "}" #define T_END_BLOCK_1 "" #define T_END_BLOCK_2 "" #define T_END_BLOCK_3 ""
				tokenSEMICOLON = ";" >> BOUNDARIES;	tokenSEMICOLON = ";" >> BOUNDARIES;	#define T_SEMICOLON_0 ";" #define T_SEMICOLON_1 "" #define T_SEMICOLON_2 "" #define T_SEMICOLON_3 ""
				tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	#define T_DATA_TYPE_0 "INTEGER16" #define T_DATA_TYPE_1 "" #define T_DATA_TYPE_2 "" #define T_DATA_TYPE_3 ""
				tokenCOMMA = "," >> BOUNDARIES;	tokenCOMMA = "," >> BOUNDARIES;	#define T_COMA_0 "," #define T_COMA_1 "" #define T_COMA_2 "" #define T_COMA_3 ""
						#define T_BITWISE_NOT_0 "~" #define T_BITWISE_NOT_1 "" #define T_BITWISE_NOT_2 "" #define T_BITWISE_NOT_3 ""
				tokenNOT = "NOT" >> STRICT_BOUNDARIES;	tokenNOT = "NOT" >> STRICT_BOUNDARIES;	#define T_NOT_0 "NOT" #define T_NOT_1 "" #define T_NOT_2 "" #define T_NOT_3 ""
						#define T_BITWISE_AND_0 "&" #define T_BITWISE_AND_1 "" #define T_BITWISE_AND_2 "" #define T_BITWISE_AND_3 ""
				tokenAND = "AND" >> STRICT_BOUNDARIES;	tokenAND = "AND" >> STRICT_BOUNDARIES;	#define T_AND_0 "AND" #define T_AND_1 "" #define T_AND_2 "" #define T_AND_3 ""
						#define T_BITWISE_OR_0 " " #define T_BITWISE_OR_1 "" #define T_BITWISE_OR_2 "" #define T_BITWISE_OR_3 ""
				tokenOR = "OR" >> STRICT_BOUNDARIES;	tokenOR = "OR" >> STRICT_BOUNDARIES;	#define T_OR_0 "OR" #define T_OR_1 "" #define T_OR_2 "" #define T_OR_3 ""
				tokenEQUAL = "==" >> BOUNDARIES;	tokenEQUAL = "==" >> BOUNDARIES;	#define T_EQUAL_0 "==" #define T_EQUAL_1 "" #define T_EQUAL_2 "" #define T_EQUAL_3 ""
				tokenNOTEQUAL = "!=" >> BOUNDARIES;	tokenNOTEQUAL = "!=" >> BOUNDARIES;	#define T_NOT_EQUAL_0 "!=" #define T_NOT_EQUAL_1 "" #define T_NOT_EQUAL_2 "" #define T_NOT_EQUAL_3 ""
				tokenLESSOREQUAL = "<=" >> BOUNDARIES;	tokenLESSOREQUAL = "<=" >> BOUNDARIES;	#define T_LESS_OR_EQUAL_0 "<=" #define T_LESS_OR_EQUAL_1 "" #define T_LESS_OR_EQUAL_2 "" #define T_LESS_OR_EQUAL_3 ""
				tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	#define T_GREATER_OR_EQUAL_0 ">=" #define T_GREATER_OR_EQUAL_1 "" #define T_GREATER_OR_EQUAL_2 "" #define T_GREATER_OR_EQUAL_3 ""
				tokenPLUS = "+" >> BOUNDARIES;	tokenPLUS = "+" >> BOUNDARIES;	#define T_ADD_0 "+" #define T_ADD_1 "" #define T_ADD_2 "" #define T_ADD_3 ""
				tokenMINUS = "-" >> BOUNDARIES;	tokenMINUS = "-" >> BOUNDARIES;	#define T_SUB_0 "-" #define T_SUB_1 "" #define T_SUB_2 "" #define T_SUB_3 ""
				tokenMUL = "*" >> BOUNDARIES;	tokenMUL = "*" >> BOUNDARIES;	#define T_MUL_0 "" #define T_MUL_1 "" #define T_MUL_2 "" #define T_MUL_3 ""
				tokenDIV = "DIV" >> STRICT_BOUNDARIES;	tokenDIV = "DIV" >> STRICT_BOUNDARIES;	#define T_DIV_0 "DIV" #define T_DIV_1 "" #define T_DIV_2 "" #define T_DIV_3 ""
				tokenMOD = "MOD" >> STRICT_BOUNDARIES;	tokenMOD = "MOD" >> STRICT_BOUNDARIES;	#define T_MOD_0 "MOD" #define T_MOD_1 "" #define T_MOD_2 "" #define T_MOD_3 ""
				tokenLRASSIGN = "=:;" >> BOUNDARIES;	tokenLRASSIGN = "=:;" >> BOUNDARIES;	#define T_LRASSIGN_0 "=:;" #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""
						#define T_ELSE_IF_0 T_ELSE_BLOCK_0 #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""
				tokenDO = "DO" >> STRICT_BOUNDARIES;	tokenDO = "DO" >> STRICT_BOUNDARIES;	#define T_DO_0 "DO" #define T_DO_1 "" #define T_DO_2 "" #define T_DO_3 ""
				tokenFOR = "FOR" >> STRICT_BOUNDARIES;	tokenFOR = "FOR" >> STRICT_BOUNDARIES;	#define T_FOR_0 "FOR" #define T_FOR_1 "" #define T_FOR_2 ""

						#define T_FOR_3 ""
						#define T_DOWNT0_0 "DOWNT0" #define T_DOWNT0_1 "" #define T_DOWNT0_2 "" #define T_DOWNT0_3 ""
				tokenDOWNT0 = "DOWNT0" >> STRICT_BOUNDARIES;	tokenDOWNT0 = "DOWNT0" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 "" #define T_INPUT_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME" #define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
						#define GRAMMAR_LL2_2025 {\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident		program_name = SAME_RULE(ident);	{ LA_IS, ("ident_terminal"), { "program_name", {\
				program_name = SAME_RULE(ident);		{ LA_IS, (""), 1, ("ident")}\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	value_type = SAME_RULE(tokenINTEGER16);	}};\
	array_specify = "[", unsigned_value, "]";	array_specify → "[", unsigned_value "]"	array_specify(1: "[" → "[", unsigned_value "]"		array_specify = "[" >> unsigned_value >> "]";	{ LA_IS, ("["), { "array_specify", {\
						{ LA_IS, (""), 3, ("[" , "unsigned_value", ""]")}\
declaration_element = ident, array_specify__optional;	declaration_element = ident, array_specify__optional;	declaration_element → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);	declaration_element = ident >> array_specify__optional;	}};\
	array_specify__optional = array_specify ε;	array_specify__optional → array_specify array_specify__optional → ε	array_specify__optional(1: "[" → array_specify array_specify__optional(1: !"[" → ε		array_specify__optional = array_specify "";	{ LA_IS, ("["), { "array_specify__optional", {\
						{ LA_IS, (""), 1, ("array_specify")}\
other_declaration_ident = ",", declaration_element;	other_declaration_ident = ",", declaration_element;	other_declaration_ident → ",", declaration_element	other_declaration_ident(1: ",") → ",", declaration_element	other_declaration_ident = tokenCOMMA >> declaration_element;	other_declaration_ident = tokenCOMMA >> declaration_element;	{ LA_NOT, ("["), { "array_specify__optional", {\
						{ LA_IS, (""), 0, ("")}\
declaration = value_type, declaration_element, other_declaration_ident_iteration;	declaration = value_type, declaration_element, other_declaration_ident_iteration;	declaration → value_type declaration_element other_declaration_ident_iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident_iteration	declaration = value_type >> declaration_element >> *other_declaration_ident;	declaration = value_type >> declaration_element >> other_declaration_ident_iteration;	}};\
	other_declaration_ident_iteration = other_declaration_ident, other_declaration_ident_iteration ε;	other_declaration_ident_iteration → other_declaration_ident other_declaration_ident_iteration false_cond_block_without_else_iteration → ε	other_declaration_ident_iteration(1: ",") → other_declaration_ident other_declaration_ident_iteration false_cond_block_without_else_iteration(1: !" ,") → ε		other_declaration_ident_iteration = other_declaration_ident >> other_declaration_ident_iteration "";	{ LA_IS, (T_COMA_0), { "other_declaration_ident", {\
						{ LA_IS, (""), 2, (T_COMA_0, "declaration_element")}\
index_action = "[", expression, "]";	index_action = "[", expression, "]";	index_action → "[" expression "]"	index_action(1: "[" → "[" expression "]"	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	}};\
unary_operator = "NOT";	unary_operator = "NOT";	unary_operator → "NOT"	unary_operator(1: "NOT") → "NOT"	unary_operator = SAME_RULE(tokenNOT);	unary_operator = SAME_RULE(tokenNOT);	{ LA_IS, (T_NOT_0), { "unary_operator", {\
						{ LA_IS, (""), 1, { T_NOT_0 }}\
unary_operation = unary_operator, expression;	unary_operation = unary_operator, expression;	unary_operation → unary_operator expression	unary_operation(1: "NOT") → unary_operator expression	unary_operation = unary_operator >> expression;	unary_operation = unary_operator >> expression;	}};\
						{ LA_IS, (T_NOT_0), { "unary_operation", {\
						{ LA_IS, (""), 2, { "unary_operator", "expression" }}\
binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "==" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "==" binary_operator → "!=" binary_operator → "<=" binary_operator → ">=" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "==") → "==" binary_operator(1: "!=") → "!=" binary_operator(1: "<=") → "<=" binary_operator(1: ">=") → ">=" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "*") → "*" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, (T_AND_0), { "binary_operator", {\
						{ LA_IS, (""), 1, { T_AND_0 }}\
						}};\
						{ LA_IS, (T_OR_0), { "binary_operator", {\
						{ LA_IS, (""), 1, { T_OR_0 }}\
						}};\
						{ LA_IS, (T_EQUAL_0), { "binary_operator", {\
						{ LA_IS, (""), 1, { T_EQUAL_0 }}\
						}};\
						{ LA_IS, (T_NOT_EQUAL_0), { "binary_operator", {\
						{ LA_IS, (""), 1, { T_NOT_EQUAL_0 }}\
						}};\
						{ LA_IS, (T_LESS_OR_EQUAL_0), { "binary_operator", {\
						{ LA_IS, (""), 1, { T_LESS_OR_EQUAL_0 }}\
						}};\
						{ LA_IS, (T_GREATER_OR_EQUAL_0), {
						"binary_operator", {\
						{ LA_IS, (""), 1, { T_GREATER_OR_EQUAL_0 }}\
						}};\
						{ LA_IS, (T_ADD_0), { "binary_operator", {\
						{ LA_IS, (""), 1, { T_ADD_0 }}\

			body_for_false__optional		body_for_false__optional;	"false_cond_block_without_else__iteration", "body_for_false__optional" }}\ }};\
	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration → ε	false_cond_block_without_else__iteration(1: "ELSE"; 2: "IF") → false_cond_block_without_else false_cond_block_without_else__iteration false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → ε false_cond_block_without_else__iteration(1: !"ELSE") → ε		false_cond_block_without_else__iteration = false_cond_block_without_else >> false_cond_block_without_else__iteration "";	{LA_IS, { T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration",{\ {LA_IS, {T_ELSE_IF_1}, 2, { "false_cond_block_without_else", "false_cond_block_without_else__iteration" }}\} {LA_NOT, { T_ELSE_IF_1 }, 0, { "" }}\} }};\} {LA_NOT, { T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration",{\ {LA_IS, {""}, 0, { "" }}\} }};\}
	body_for_false__optional = body_for_false ε;	body_for_false__optional → body_for_false body_for_false__optional → ε	body_for_false__optional(1: "FALSE") → body_for_false body_for_false__optional(1: !"FALSE") → ε		body_for_false__optional = body_for_false "";	{LA_IS, { T_ELSE_BLOCK_0 }, { "body_for_false__optional",{\ {LA_IS, {""}, 1, { "body_for_false" }}\} }};\} {LA_NOT, { T_ELSE_BLOCK_0 }, { "body_for_false__optional",{\ {LA_IS, {""}, 0, { "" }}\} }};\}
cycle_begin_expression = expression;	cycle_begin_expression = expression;	cycle_begin_expression → expression	cycle_begin_expression(1: "(" , "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	cycle_begin_expression = SAME_RULE(expression);	cycle_begin_expression = SAME_RULE(expression);	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_begin_expression",{\ {LA_IS, {""}, 1, { "expression" }}\} }};\}
cycle_end_expression = expression;	cycle_end_expression = expression;	cycle_end_expression → expression	cycle_end_expression(1: "(" , "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	cycle_end_expression = SAME_RULE(expression);	cycle_end_expression = SAME_RULE(expression);	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_end_expression",{\ {LA_IS, {""}, 1, { "expression" }}\} }};\}
cycle_counter = ident;	cycle_counter = ident;	cycle_counter → ident	cycle_counter(1: "_") → ident	cycle_counter = SAME_RULE(ident);	cycle_counter = SAME_RULE(ident);	{LA_IS, { "ident_terminal" }, { "cycle_counter",{\ {LA_IS, {""}, 1, { "ident" }}\} }};\}
cycle_counter_lr_init = cycle_begin_expression , "=" , cycle_counter;	cycle_counter_lr_init = cycle_begin_expression , "=" , cycle_counter;	cycle_counter_lr_init → cycle_begin_expression "=" cycle_counter	cycle_counter_lr_init(1: "(" , "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → cycle_begin_expression "!=" cycle_counter	cycle_counter_lr_init = cycle_begin_expression >> tokenLRASSIGN >> cycle_counter;	cycle_counter_lr_init = cycle_begin_expression >> tokenLRASSIGN >> cycle_counter;	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_lr_init",{\ {LA_IS, {""}, 3, { "cycle_begin_expression", T_LRASSIGN_0, "cycle_counter" }}\} }};\}
cycle_counter_init = cycle_counter_lr_init;	cycle_counter_init = cycle_counter_lr_init;	cycle_counter_init → cycle_counter_lr_init	cycle_counter_init(1: "(" , "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → cycle_counter_lr_init	cycle_counter_init = SAME_RULE(cycle_counter_lr_init);	cycle_counter_init = SAME_RULE(cycle_counter_lr_init);	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_init",{\ {LA_IS, {""}, 1, { "cycle_counter_lr_init" }}\} }};\}
cycle_counter_last_value = cycle_end_expression;	cycle_counter_last_value = cycle_end_expression;	cycle_counter_last_value → cycle_end_expression	cycle_counter_last_value(1: "(" , "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → cycle_end_expression	cycle_counter_last_value = SAME_RULE(cycle_end_expression);	cycle_counter_last_value = SAME_RULE(cycle_end_expression);	{LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "cycle_counter_last_value",{\ {LA_IS, {""}, 1, { "cycle_end_expression" }}\} }};\}
cycle_body = "DO" , (statement block_statements);	cycle_body = "DO" , statement__or__block_statements;	cycle_body → "DO" statement__or__block_statements	cycle_body(1: "DO") → "DO" statement__or__block_statements	cycle_body = tokenDO >> (statement block_statements);	cycle_body = tokenDO >> statement__or__block_statements;	{LA_IS, { T_DO_0 }, { "cycle_body",{\ {LA_IS, {""}, 2, { T_DO_0, "statement__or__block_statements" }}\} }};\}
fordownto_cycle = "FOR" , cycle_counter_init , "DOWNT0" , cycle_counter_last_value , cycle_body;	fordownto_cycle = "FOR" , cycle_counter_init , "DOWNT0" , cycle_counter_last_value , cycle_body;	fordownto_cycle → "FOR" cycle_counter_init "DOWNT0" , cycle_counter_last_value cycle_body	fordownto_cycle(1: "FOR") → "FOR" cycle_counter_init "DOWNT0", cycle_counter_last_value cycle_body	fordownto_cycle = tokenFOR >> cycle_counter_init >> tokenDOWNT0 >> cycle_counter_last_value >> cycle_body;	fordownto_cycle = tokenFOR >> cycle_counter_init >> tokenDOWNT0 >> cycle_counter_last_value >> cycle_body;	{LA_IS, { T_FOR_0 }, { "fordownto_cycle",{\ {LA_IS, {""}, 5, { T_FOR_0, "cycle_counter_init", T_DOWNT0_0, "cycle_counter_last_value", "cycle_body" }}\} }};\}
	statement__or__block_statements = statement block_statements;	statement__or__block_statements → statement block_statements	statement__or__block_statements(1: "(") → statement statement__or__block_statements(1: "(") → block_statements		statement__or__block_statements = statement block_statements;	{LA_IS, { "ident_terminal", "(" , T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__or__block_statements",{\ {LA_IS, {""}, 1, { "statement" }}\} }};\} {LA_IS, { T_BEGIN_BLOCK_0 }, { "statement__or__block_statements",{\ {LA_IS, {""}, 1, { "block_statements" }}\} }};\}
input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input_rule = "GET" , argument_for_input;	input_rule → "GET" argument_for_input	input_rule(1: "GET") → "GET" argument_for_input	input_rule = tokenGET >> (ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >> tokenGROUPEXPRESSIONEND);	input_rule = tokenGET >> argument_for_input;	{LA_IS, { T_INPUT_0 }, { "input_rule",{\ {LA_IS, {""}, 2, { T_INPUT_0, "argument_for_input" }}\} }};\}
	argument_for_input = ident , index_action__optional; argument_for_input = "(" , "ident", "index_action__optional", ")" ;	argument_for_input → ident index_action__optional argument_for_input → "(" "ident" "index_action__optional" ")"	argument_for_input(1: "_") → ident index_action__optional argument_for_input(1: "(") → "(" "ident" "index_action__optional" ")"		argument_for_input = ident >> index_action__optional tokenGROUPEXPRESSIONBEGIN >> ident >> index_action__optional >> tokenGROUPEXPRESSIONEND;	{LA_IS, { "ident_terminal" }, { "argument_for_input",{\ {LA_IS, {""}, 2, { "ident", "index_action__optional" }}\} }};\} {LA_IS, { "("), { "argument_for_input",{\ {LA_IS, {""}, 4, { "(" , "ident", "index_action__optional", ")" }}\} }};\}
output_rule = "PUT" , expression;	output_rule = "PUT" , expression;	output_rule → "PUT" expression	output(1: "PUT") → "PUT" expression	output_rule = tokenPUT >> expression;	output_rule = tokenPUT >> expression;	{LA_IS, { T_OUTPUT_0 }, { "output_rule", {\ {LA_IS, {""}, 2, { T_OUTPUT_0, "expression" }}\} }};\}
statement = expression_or_cond_block__with_optional_assign fordownto_cycle input_rule output_rule ";";	statement = expression_or_cond_block__with_optional_assign fordownto_cycle input_rule output_rule ";";	statement → expression_or_cond_block__with_optional_assign statement → fordownto_cycle statement → input_rule statement → output_rule statement → ";"	statement(1: "(" , "NOT", "+", "-", "_", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression_or_cond_block__with_optional_assign statement(1: "FOR") → fordownto_cycle statement(1: "GET") → input_rule statement(1: "PUT") → output_rule statement(1: ";") → ";"	statement = expression_or_cond_block__with_optional_assign fordownto_cycle input_rule output_rule tokenSEMICOLON;	statement = expression_or_cond_block__with_optional_assign fordownto_cycle input_rule output_rule tokenSEMICOLON;	{LA_IS, { "(" , T_NOT_0, "ident_terminal", "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0 }, { "statement", {\ { LA_IS, {""}, 1, {"expression_or_cond_block__with_optional_assign"}}\} }};\} {LA_IS, { T_FOR_0 }, { "statement", {\ {LA_IS, {""}, 1, {"fordownto_cycle"}}\} }};\} {LA_IS, { T_INPUT_0 }, { "statement", {\ {LA_IS, {""}, 1, {"input_rule"}}\} }};\} {LA_IS, { T_OUTPUT_0 }, { "statement", {\ {LA_IS, {""}, 1, {"output_rule"}}\} }};\} {LA_IS, { T_SEMICOLON_0 }, { "statement", {\ {LA_IS, {""}, 1, {";"}}\} }};\}
	statement__iteration = statement, statement__iteration ε;	statement__iteration → statement statement__iteration statement__iteration → ε	statement__iteration(1: "_" , "(" , "NOT", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "+", "-", "IF", "FOR", "GET", "PUT", ";") → statement statement__iteration statement__iteration(1: !"_" , !"(", !"NOT", !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"+", !"-", !"IF", !"FOR", !"GET", !"PUT", !";") → "		statement__iteration = statement >> statement__iteration "";	{ LA_IS, { "ident_terminal", "(" , T_NOT_0, "unsigned_value_terminal", T_ADD_0, T_SUB_0, T_IF_0, T_FOR_0, T_INPUT_0, T_OUTPUT_0, T_SEMICOLON_0 }, { "statement__iteration",{\ { LA_IS, {""}, 2, { "statement", "statement__iteration" }}\}

				tokenPUT tokenNAME tokenBODY tokenDATA tokenBEGIN tokenEND tokenBEGINBLOCK tokenENDBLOCK tokenLEFTSQUAREBRACKETS tokenRIGHTSQUAREBRACKETS tokenSEMICOLON) >> tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;	tokenPUT tokenNAME tokenBODY tokenDATA tokenBEGIN tokenEND tokenBEGINBLOCK tokenENDBLOCK tokenLEFTSQUAREBRACKETS tokenRIGHTSQUAREBRACKETS tokenSEMICOLON) >> tokenUNDERSCORE >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> letter_in_upper_case >> STRICT_BOUNDARIES;	
letter_in_lower_case = "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u" "v" "w" "x" "y" "z";	letter_in_lower_case → "a" letter_in_lower_case → "b" letter_in_lower_case → "c" letter_in_lower_case → "d" letter_in_lower_case → "e" letter_in_lower_case → "f" letter_in_lower_case → "g" letter_in_lower_case → "h" letter_in_lower_case → "i" letter_in_lower_case → "j" letter_in_lower_case → "k" letter_in_lower_case → "l" letter_in_lower_case → "m" letter_in_lower_case → "n" letter_in_lower_case → "o" letter_in_lower_case → "p" letter_in_lower_case → "q" letter_in_lower_case → "r" letter_in_lower_case → "s" letter_in_lower_case → "t" letter_in_lower_case → "u" letter_in_lower_case → "v" letter_in_lower_case → "w" letter_in_lower_case → "x" letter_in_lower_case → "y" letter_in_lower_case → "z"	letter_in_lower_case(1: "a") → "a" letter_in_lower_case(1: "b") → "b" letter_in_lower_case(1: "c") → "c" letter_in_lower_case(1: "d") → "d" letter_in_lower_case(1: "e") → "e" letter_in_lower_case(1: "f") → "f" letter_in_lower_case(1: "g") → "g" letter_in_lower_case(1: "h") → "h" letter_in_lower_case(1: "i") → "i" letter_in_lower_case(1: "j") → "j" letter_in_lower_case(1: "k") → "k" letter_in_lower_case(1: "l") → "l" letter_in_lower_case(1: "m") → "m" letter_in_lower_case(1: "n") → "n" letter_in_lower_case(1: "o") → "o" letter_in_lower_case(1: "p") → "p" letter_in_lower_case(1: "q") → "q" letter_in_lower_case(1: "r") → "r" letter_in_lower_case(1: "s") → "s" letter_in_lower_case(1: "t") → "t" letter_in_lower_case(1: "u") → "u" letter_in_lower_case(1: "v") → "v" letter_in_lower_case(1: "w") → "w" letter_in_lower_case(1: "x") → "x" letter_in_lower_case(1: "y") → "y" letter_in_lower_case(1: "z") → "z"	A = "A"; B = "B"; C = "C"; D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	A = "A"; B = "B"; C = "C"; D = "D"; E = "E"; F = "F"; G = "G"; H = "H"; I = "I"; J = "J"; K = "K"; L = "L"; M = "M"; N = "N"; O = "O"; P = "P"; Q = "Q"; R = "R"; S = "S"; T = "T"; U = "U"; V = "V"; W = "W"; X = "X"; Y = "Y"; Z = "Z"; letter_in_lower_case = a b c d e f g h i j k l m n o p q r s t u v w x y z;	\	
letter_in_upper_case = "A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K" "L" "M" "N" "O" "P" "Q" "R" "S" "T" "U" "V" "W" "X" "Y" "Z";	letter_in_upper_case → "A" letter_in_upper_case → "B" letter_in_upper_case → "C" letter_in_upper_case → "D" letter_in_upper_case → "E" letter_in_upper_case → "F" letter_in_upper_case → "G" letter_in_upper_case → "H" letter_in_upper_case → "I" letter_in_upper_case → "J" letter_in_upper_case → "K" letter_in_upper_case → "L" letter_in_upper_case → "M" letter_in_upper_case → "N" letter_in_upper_case → "O" letter_in_upper_case → "P" letter_in_upper_case → "Q" letter_in_upper_case → "R" letter_in_upper_case → "S" letter_in_upper_case → "T" letter_in_upper_case → "U" letter_in_upper_case → "V" letter_in_upper_case → "W" letter_in_upper_case → "X" letter_in_upper_case → "Y" letter_in_upper_case → "Z"	letter_in_upper_case(1: "A") → "A" letter_in_upper_case(1: "B") → "B" letter_in_upper_case(1: "C") → "C" letter_in_upper_case(1: "D") → "D" letter_in_upper_case(1: "E") → "E" letter_in_upper_case(1: "F") → "F" letter_in_upper_case(1: "G") → "G" letter_in_upper_case(1: "H") → "H" letter_in_upper_case(1: "I") → "I" letter_in_upper_case(1: "J") → "J" letter_in_upper_case(1: "K") → "K" letter_in_upper_case(1: "L") → "L" letter_in_upper_case(1: "M") → "M" letter_in_upper_case(1: "N") → "N" letter_in_upper_case(1: "O") → "O" letter_in_upper_case(1: "P") → "P" letter_in_upper_case(1: "Q") → "Q" letter_in_upper_case(1: "R") → "R" letter_in_upper_case(1: "S") → "S" letter_in_upper_case(1: "T") → "T" letter_in_upper_case(1: "U") → "U" letter_in_upper_case(1: "V") → "V" letter_in_upper_case(1: "W") → "W" letter_in_upper_case(1: "X") → "X" letter_in_upper_case(1: "Y") → "Y" letter_in_upper_case(1: "Z") → "Z"	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	a = "a"; b = "b"; c = "c"; d = "d"; e = "e"; f = "f"; g = "g"; h = "h"; i = "i"; j = "j"; k = "k"; l = "l"; m = "m"; n = "n"; o = "o"; p = "p"; q = "q"; r = "r"; s = "s"; t = "t"; u = "u"; v = "v"; w = "w"; x = "x"; y = "y"; z = "z"; letter_in_upper_case = A B C D E F G H I J K L M N O P Q R S T U V W X Y Z;	\	
				STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) (!(qi::alpha qi::char_("_ "))); BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY); BOUNDARY = BOUNDARY__SPACE BOUNDARY__TAB BOUNDARY__VERTICAL_TAB BOUNDARY__FORM_FEED BOUNDARY__CARRIAGE_RETURN BOUNDARY__LINE_FEED BOUNDARY__NULL; BOUNDARY__SPACE = " "; BOUNDARY__TAB = "\t"; BOUNDARY__VERTICAL_TAB = "\v"; BOUNDARY__FORM_FEED = "\f"; BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	STRICT_BOUNDARIES = (BOUNDARY >> *(BOUNDARY)) (!(qi::alpha qi::char_("_ "))); BOUNDARIES = (BOUNDARY >> *(BOUNDARY) NO_BOUNDARY); BOUNDARY = BOUNDARY__SPACE BOUNDARY__TAB BOUNDARY__VERTICAL_TAB BOUNDARY__FORM_FEED BOUNDARY__CARRIAGE_RETURN BOUNDARY__LINE_FEED BOUNDARY__NULL; BOUNDARY__SPACE = " "; BOUNDARY__TAB = "\t"; BOUNDARY__VERTICAL_TAB = "\v"; BOUNDARY__FORM_FEED = "\f"; BOUNDARY__CARRIAGE_RETURN = "\r"; BOUNDARY__LINE_FEED = "\n"; BOUNDARY__NULL = "\0"; NO_BOUNDARY = ""; #define WHITESPACES \ STRICT_BOUNDARIES, \ BOUNDARIES, \ BOUNDARY, \ BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	\
						\ \ \ { LA_IS, { T_NAME_0 }, { "program____part1", \ { LA_IS, { "" }, 7, { T_NAME_0, "program_name",

						T_SEMICOLON_0, T_BODY_0, T_DATA_0, "declaration_optional", T_SEMICOLON_0 }}\ }}\ \ }.\ "program_rule"
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